

Yuxian Li

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Interested in how to steer AI to continuously learn from messy, heterogeneous physical-world data, and designing the low-level compute infrastructure to scale it.

EDUCATION

The University of Hong Kong

B.Eng. in Computer Engineering



Expected 2027

Selected Courses: Probability & Statistics, Linear Algebra, Discrete Math, Computer Architecture, Quantum Computing
Outstanding Poster Award, HKU ECE Poster Display Day, 2026 (1 of 13 recipients in a 158-student graduating cohort)

University of California, Berkeley

Visiting Student. Data Structures and Algorithms

Summer 2023

RESEARCH EXPERIENCE

CAFE: Counterfactual Benchmark for Segmentation Models^[1] [arXiv](#) [Project Page](#)

2026

Agent Systems and Evaluation Lead, mentored by Profs. Xihui Liu (HKU MMLab) and Han Wang

- Even strong segmentation models can draw the right outline but identify the wrong object or concept. So I discussed these issues with my coauthors to formulate the research problem, and together we built CAFE to expose and quantify them.
- Co-curated 2,146 cases from 48,423 AI-edited candidates through three-stage expert review, checking edit quality while retaining the target's location and outline. We evaluated six segmentation systems (e.g., end-to-end, modular, and agentic) with correct and misleading prompts across 4,292 episodes.
- Conceived and built CAFE-SAM3, a GPT-5.5 visual agent that revises prompts and zooms in on material and texture cues to select a matching mask or abstain. The agent raised class-gated F1 from 38.5 to 63.3 (+24.8 points) and reduced target-aligned concept swaps from 8.9% to 1.7% on CAFE.
- Led review-stage follow-up experiments and failure analysis to address reviewer questions. I built a fault-tolerant Ray evaluation pipeline with GPU-resident SAM 3 actors, checkpoint recovery, and traceable reruns to support these studies.

Human-Signal-Guided Representation Learning for VLA Policies

2026

Research Intern at [HKU InfoBodied AI Lab](#), mentored by Prof. Yanchao Yang, with a manuscript in preparation

- Co-developing human-signal-guided representations for robot policy learning. I contribute to temporal alignment across modalities and design learning objectives and ablation studies.

Continuous-Time ODE Solver for Flow-Matching Action Generation

2025

Research Assistant, Center for Advanced Semiconductors and Integrated Circuits (CASIC), HKU

Supervised by Prof. Han Wang. Manuscript under review.

- On-device inference latency limits how quickly learned robot policies can respond in dynamic environments. I designed circuit-compatible flow-matching action models for faster closed-loop control.
- Designed the distillation method and built training and evaluation pipelines, with ablations across four [LIBERO](#) suites and real-robot tasks.
- Designed three language-conditioned tasks and built the robot-camera setup, including custom 3D-printed fixtures and teleoperated data collection.
- Collaborated closely with circuit and chip designers on model-hardware co-design, including noise- and quantization-aware training.

PUBLICATIONS

^[1] **From Pixels to Concepts: Do Segmentation Models Understand What They Segment?**

S. Liang*, Z. Wang*, Y. Li*, X. Liu, and H. Wang

NeurIPS 2026 Evaluations & Datasets Track (Poster)

*Equal contribution (co-first authors).

INDUSTRY EXPERIENCE

OneRobotics (HKEX: 6600)

Research Intern, Multimodal Foundation Models

Aug to Sep 2026

Shenzhen, China

- Driving online RL for humanoid loco-manipulation with VLA and world-action models (WAMs) for reliable household deployment. I am co-developing a closed-loop RL pipeline that uses rollout feedback to update and redeploy policies.
- Developing and scaling multimodal pretraining pipelines for VLA and WAMs. I work with colleagues to refine episode/subtask labeling, cross-embodiment retargeting, and UMI/egocentric data mixing.

Inspur Group

Software Development Intern

Sep to Dec 2024

Jinan, China

- Built front-end workflows for an enterprise platform and worked with backend engineers to integrate and test APIs against business requirements.

OPEN SOURCE

Scholar-MCP [GitHub](#)

Lead Developer and Maintainer

2025 to Present

- An MCP server that helps agents find papers and reuse findings in auto-research workflows. I built parallel multi-source search and reranking to improve retrieval efficiency and limit source bias, raising top-five query hit rate from 32% to 62% on a fixed 50-query [LitSearch](#) sample.
- Designed tools for agents to refine queries and follow citations, with source-linked, length-bounded responses to steer long-running research. A searchable SQLite-backed library preserves papers and notes across sessions, reducing repeated searches and context overhead.

LeRobot 🤖 [GitHub](#)

Upstream Contributor, Hugging Face

2026

- RealSense D405 startup failures previously required a hard manual USB reset, so I added automatic recovery in merged [PR #3894](#) with thread-safe frame handling and exception-safe cleanup to prevent stale-frame writes during restarts and improve startup reliability.
- To make robot policies more robust to real-world visual variation, I implemented configurable PyTorch augmentations and integrated them into LeRobot's torchvision v2 pipeline via [PR #4210](#) and released [robo-augment](#) as a reusable library.

Reveal StarMap [GitHub](#) [Steam Workshop](#)

Lead Developer and Maintainer

2026

- A C# mod for the game Oxygen Not Included that lets players preview the full star map without changing saved exploration progress. I reverse-engineered Unity's map and camera behavior and used Harmony patches to preserve and restore exploration and camera state.
- Handled camera bounds across scene changes and save loading, and restored state after mod conflicts. I published it on Steam Workshop, gaining thousands of subscribers.

MediBot [GitHub](#)

Project Lead (3-person team)

2024 to 2025

- Led end-to-end development of an autonomous mobile robot for contactless hospital patrols during COVID-19, integrating a Jetson Nano and mecanum base with Arduino motor control and ESP32 telemetry.
- To support autonomous patrol, I published encoder odometry and TF in ROS and configured YDLidar navigation with GMapping, AMCL localization, NavFn global planning, DWA local planning, and layered costmaps, producing a lab map with 5 cm resolution.
- Integrated thermal imaging and camera-based face and mask screening. I built a remote interface with live camera and LiDAR views, MQTT telemetry and motion commands, and emergency manual takeover.

SELECTED MINI-PROJECTS

Online Mental Health Risk Predictor [GitHub](#)

Project Lead (3-person team)

2026

Parallel Jacobi Solver [GitHub](#)

2025

UniHaven Off-Campus Housing Platform [GitHub](#)

Requirements and Backend Engineer (4-person team)

2024 to 2025

Multithreaded Path-Tracing Engine

2023

Infected Utopia Terminal RPG [GitHub](#)

Game Designer, Narrative and Combat Systems (4-person team)

2023

TECHNICAL SKILLS

Agents Agentic/loop programming/engineering, agent harnesses (Pi, Claude Code, Codex), MCP

Programming & ML Python, C++, C, C#, Java; PyTorch (FSDP), DeepSpeed, Ray, vLLM

Robotics ROS, LeRobot, ALOHA (Piper, ARX), Panda

Hardware Prototyping 3D modeling, 3D printing, soldering

Languages Mandarin Chinese, English (fluent)